

Chacha Chen

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Address: College of IST, State College, Pennsylvania

RESEARCH ☐ **Applied Machine Learning** ☐ **Human Computer Interaction**
KEYWORDS ☐ Social Computing ☐ Natural Language Processing
 ☐ Healthcare ☐ Crowdsourcing

EDUCATION **Pennsylvania State University** Aug. 2019 - Dec. 2020 (expected)
College of Information Sciences and Technologies
Master in Informatics
• **Advisor:** Ting-Hao (Kenneth) Huang

Shanghai Jiao Tong University Sep. 2015 – Jun. 2019
Department of Computer Science
Bachelor of Science in Computer Science
• **Advisor:** Weinan Zhang, Zhenhui (Jessie) Li

Technical University of Munich Apr. 2017 – Sep. 2017
Exchange Student in Department of Informatics

RESEARCH & **IQVIA, Analytics Center of Excellence, Boston**
WORK Machine Learning Research Intern May. 2020–Aug. 2020
EXPERIENCE • Gaussian Process for health risk prediction: Designed Bayesian machine learning models for informative health risk prediction. Different from previous methods, our proposed model is able to give uncertainty-based rich prediction. Moreover, we leveraged multi-sourced data with different modalities.
 • Supervised by Cao (Danica) Xiao (IQVIA), Prof. Fenglong Ma (PSU) and Prof. Jimeng Sun (UIUC).

CityBrain Lab, Hangzhou Jul. 2018–Aug. 2019
Machine Learning Research Intern
• Reinforcement learning for traffic signal control: Designed and implemented a reinforcement learning (RL) approach with justification on state and reward design for multi-intersection traffic signal control
• Supervised by Prof Zhenhui (Jessie) Li (PSU).

SJTU PAMI Lab, Shanghai Sep. 2017–Mar. 2018
Research Assistant
• Implementation of SVM with indefinite and semi-definite kernels: Coauthored a software package which accommodates Support Vector Machine (SVM) classification algorithms with indefinite kernels and semi-definite kernels. Moreover, we incorporated Sequential Minimal Optimization (SMO) in the traditional SVM algorithm and Provided various kernel functions, including TAHN, TL1, gaussian, polynomial and linear kernels
• Supervised by Prof Xiaolin Huang (SJTU).

PUBLICATIONS **Preprints**

1. Tongan Cai*, **Chacha Chen***, Ting-Hao K. Huang, Frank E. Ritter
Quantitative Analysis of Bibliography Management Applications
To be submitted to *CHI Late Breaking Work 2021*.
2. Jiaqi Wang, Hua Shen, **Chacha Chen**, Frank E. Ritter
Toward an Interactive Online Course Framework
Submitted to *CHI Case Studies 2021*, under review.
3. **Chacha Chen**, Junjie Liang, Fenglong Ma, Lucas Glass, Jimeng Sun, Cao Xiao
UNITE: Uncertainty-based Health Risk Prediction Leveraging Multi-sourced Data
Submitted to *WWW 2021*, under review.
4. Guanjie Zheng, Chang Liu, Hua Wei, Porter Jenkins, **Chacha Chen**, Zhenhui Li
A Residual Model on Various Tasks
Submitted to *AAAI 2021*, under review.

Conference Papers

1. Guanjie Zheng, Chang Liu, Hua Wei, **Chacha Chen**, Zhenhui Li, Rebuilding City-Wide Traffic from Multi-Source Data. In Proceedings of 37th IEEE International Conference on Data Engineering (**ICDE 2021**), Chania, Greece, April 2021.
2. Hua Wei, **Chacha Chen**, Chang Liu, Guanjie Zheng and Zhenhui Li, ImIn-GAIL: Learning to Simulate with Imitation-Interpolation on Sparse Trajectory Data. In Proceedings of European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD 2020**), Ghent, Belgium, Sep 2020. (**Best Applied Data Science Paper Award**)
3. **Chacha Chen**, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu and Zhenhui Li, Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control. In Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**), New York, United States, February 2020. (Acceptance rate: $1591/7737=20.6\%$)
4. Hua Wei, Nan Xu, Huichu Zhang, Guanjie Zheng, Xinshi Zang, **Chacha Chen**, Weinan Zhang, Yanmin Zhu, Kai Xu and Zhenhui Li, CoLight: Learning Network-level Cooperation for Traffic Signal Control. In Proceedings of the 2019 ACM on Conference on Information and Knowledge Management (**CIKM 2019**), Beijing, China, November 2019. (Acceptance rate: $200/1030=19.4\%$)
5. Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2019**), Anchorage, AK, USA, Aug 2019. (Acceptance rate: $170/1200=14.2\%$)

Workshop Papers

1. **Chacha Chen**, Guanjie Zheng, Hua Wei, Zhenhui Li, Physics-informed Generative Adversarial Networks. In Proceedings of the NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning (**IIBPSL, NeurIPS 2020**).
2. **Chacha Chen**, Chieh-Yang Huang, Yaqi Hou, Yang Shi, Enyan Dai, Jiaqi Wang, Joint Event Multi-task Learning for Slot Filling in Noisy Text. In Proceedings of the Workshop on Noisy User-generated Text (**W-NUT, EMNLP 2020**).
3. Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, Deep Reinforcement Learning for Traffic Signal Control along Arterials. In Proceedings of the Workshop on Deep Reinforcement Learning for Knowledge Discovery (**DRL4KDD, KDD 2019**).

COMMUNITY ACTIVITIES & SERVICE	Conference Program Committee Member/Reviewer	
	• AAAI 2021: The Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI-21) • W-NUT: The 6th Workshop on Noisy User-generated Text with EMNLP 2020 • NeurIPS workshop on Interpretable Inductive Biases and Physically Structured Learning	
	Journal Invited Reviewer	
	• ACM Transaction on Intelligent System and Technology (TIST)	
VOLUNTEERS	AAAI-21 Student Technical Volunteer	Nov. 2020–Feb. 2021
	Developing the AAAI-21 virtual website	
	EMNLP-20 Student Volunteer	Nov. 2020
	Helpdesk	
	AAAI-20 Student Volunteer	Feb. 2020
	Poster session set-up	
TEACHING EXPERIENCE	Pennsylvania State University	
	Teaching Assistant	Fall 2020
	IST 261 (Java Course): Application Development Design Studio I	
	Prof. Kenneth Huang	
AWARDS	AAAI-20 VISA Research Scholarship	Feb. 2020
	AAAI-20 Student Travel Grant	Feb. 2020
	Top 25 in the finalist, <i>KDD Cup 2020 LDR Competition</i>	Aug. 2020
	Distinguished Bachelor Thesis, <i>Shanghai Jiao Tong University</i>	Jun. 2019
	Academic Excellence Scholarship, <i>Shanghai Jiao Tong University</i>	Sep. 2016
RELEVANT SKILLS	Technical:	Python, C/C++, MySQL, Java, Matlab, R, HTML/CSS
	Languages:	TOEFL (112, Reading 27, Listening 30, Speaking 25, Writing 30)